

Popular Belief Meets Surgical Reality: Impact of Lunar Phases, Friday the 13th and Zodiac Signs on Emergency Operations and Intraoperative Blood Loss

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Abstract

Background The influence of superstition, moon calendars, and popular belief on evidence-based medicine is stunning. More than 40% of medical staff is convinced that lunar phases can affect human behavior. The idea that Friday the 13th is associated with adverse events and bad luck is deep-rooted in the population of Western industrial countries. The aim of the present study was to test the hypothesis that these myths are transferable to real-life surgery.

Methods We analyzed the extent to which moon phases, zodiac signs, and Friday the 13th influence blood loss, emergency frequency, and intestinal perforations by evaluating the operation records of all 27,914 consecutive patients of our institution undergoing general, visceral, or vascular surgery between August 2001 and August 2010. Dates of surgery were allocated to lunar phases and to zodiac signs, as well as to Friday the 13th.

Results A total of 111 lunar cycles and 15 Fridays the 13th occurred within the 3,281-day observation period. Patients' characteristics did not differ in lunar phases, zodiac signs, or Fridays the 13th. Full moon phases, the presence of Friday the 13th, and zodiac signs influenced neither intraoperative blood loss nor emergency frequency.

No statistical peaks regarding perforated aortic aneurysms and gastrointestinal perforations were found on full moon or Friday the 13th.

Conclusions Scientific analysis of our data does not support the belief that moon phases, zodiac signs, or Friday 13th influence surgical blood loss and emergency frequency. Our data indicate that such beliefs are myths far beyond reality.

Introduction

In ancient and modern cultures, the moon is regarded as a factor influencing human fate and health. The term “lunacy” is derived from Luna, the Roman goddess of the moon, and from the belief that the power of the moon can cause disorders of the mind [1, 2]. The lunar phase and the position of the moon in the zodiac is especially claimed to be relevant for the risk of emergency operations or intraoperative complications in surgery. More than 10% of the German population believe in a relationship between lunar phase and diseases [3]. 43% of health professionals believe that lunar effects can affect human individual behavior [4]. The belief that lunar phases can influence illness may be based on Galen and Hippocrates, but no documents exist to prove that myth. In recent years many articles assessing the influence of lunar phases on suicide [5–7], mental health [8], and various accident events [9, 10] have been published. Other forms of superstition, such as signs of the zodiac may affect human behavior in various ways. A belief in Friday the 13th as an ill-fated day is deep-rooted in the population [11, 12]. To the best of our knowledge, there is only one study dealing with the zodiac and its influence on emergency cases [13].

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The aim of our retrospective study over 9 years was to evaluate if there were any statistically significant peaks of intraoperative blood loss or frequency of emergency cases that could be linked to lunar phenomena, Friday the 13th, or zodiac signs.

Patients and methods

The database was comprised of all consecutive patients admitted and undergoing general, visceral, or vascular surgery between August 29, 2001 and August 23, 2010. However, pediatric surgery and all other surgical specialities (e.g., orthopedic and trauma surgery, neurosurgery, thoracic surgery, etc.) were excluded from the evaluation. Collective operative records were evaluated using the hospital information system SAP/i.s.h.med (SAP, St. Leon-Rot, Germany). Intraoperative documentations of estimated surgical blood loss were analyzed.

One lunar phase was defined as 29.53 days (29 days, 12 h, 44 min, 2.9 s). Friday the 13th occurs when the thirteenth day of a month falls on a Friday. In the Gregorian calendar, this occurs at least once, but at most three times a year. Any month's thirteenth day will fall on a Friday whenever the month starts on a Sunday. Calculation of patient's zodiac signs, lunar phases, and zodiac signs at time of the operation, as well as identification of Friday the 13th, were performed using VBA scripts in Microsoft Excel (Microsoft, Unterschleissheim, Germany) based on the date of the operation. All values are expressed as mean $\pm$ SEM. After proving the assumption of normality and homogeneity of variance across groups, differences between the groups were calculated with the Student's *t*-test. In case of non-normal distribution of data, the Mann-Whitney *U*-test was performed. Comparison between groups of categorical data was performed with Fisher's exact test or the chi-square test. Overall statistical significance was set at $P < 0.05$. Statistical analysis was performed with SPSS 15 software package (SPSS, Chicago, IL).

Results

A total of 111 lunar phases over 3,281 days were evaluated regarding operations of 27,914 patients (16,310 men, 58.4%; 11,604 women 41.6%) with an average age of 58.8 ± 0.1 (16–99) years. At least one operation was performed at 3,153 of the 3,281 days within the observation period (96.1%). Mean estimated blood loss was 161.7 ± 2.8 ml, ranging from 1 ml to 12,000 ml. In 865 procedures (3.1%), an intraoperative blood transfusion was mandatory.

Table 1 Distribution of 27,914 operations to lunar phases and patient's characteristics from August 29 2001 to August 23 2010

Lunar phase	<i>n</i>	Percentage	Age, years
Waning moon	13,019	46.6	58.8 ± 0.1
New moon	912	3.3	58.6 ± 0.5
Full moon	916	3.3	58.6 ± 0.5
Waxing moon	13,067	46.8	58.6 ± 0.1
Overall	27,914	100.0	58.8 ± 0.1

Table 2 Emergency frequency and estimated blood loss in relation to lunar phases and Friday the 13th

Time point or lunation phase	Emergency rate (%)	Chi square	<i>P</i> value	Estimated blood loss, ml	<i>P</i> value
Friday the 13th	20.0	3.384	0.069	203.0 ± 34.8	0.245
Normal day	26.2			161.5 ± 2.8	
Friday the 13th	20.0	0.955	0.356	203.0 ± 34.8	0.079
Every other Friday	23.2			148.8 ± 5.6	
Waning moon	26.5	2.115	0.549	160.6 ± 4.0	0.676
New moon	25.8			151.98 ± 13.6	
Full moon	27.4			149.69 ± 15.4	
Waxing moon	25.8			164.40 ± 4.16	

Of all the operations analyzed, 7,309 (26.2%) were performed as emergency operations; all others were planned. Most operations were performed during waxing moon and waning moon; 912 operations were performed during a new moon, and 916 operations were performed during a full moon (Table 1). Friday the 13th occurred 15 times within the observation period, with overall 170 operations on these days. A little more than 50% (51.8%; $n = 88$) of these operations were performed during the waning moon; 48.2% ($n = 82$) were done during the waxing moon. The “worst case scenario”—a combination of a full moon and Friday the 13th—never occurred. Estimated blood loss and emergency frequency did not correlate with lunar phases, the presence of Friday the 13th (Table 2), or zodiac signs at the time of the operations (Table 3). During the zodiac signs of Capricorn (Dec 22–Jan 20), Aries (March 21–April 20), and Gemini (May 22–June 21), the rate of emergency

Table 3 Influence of zodiac signs on emergency operations and estimated intraoperative blood loss

Zodiac sign	Emergency rate	Estimated blood loss, ml
Capricorn	32.5% 610/1,877	156.3 ± 9.4
Aries	29.1% 621/2,134	168.0 ± 9.3
Gemini	28.3% 708/2,501	158.5 ± 9.3
Leo	26.6% 629/2,363	154.1 ± 9.5
Taurus	26.5% 612/2,307	149.9 ± 8.7
Cancer	26.3% 676/2,572	181.1 ± 10.8
Scorpio	25.5% 628/2,461	161.7 ± 9.1
Libra	25.3% 590/2,333	163.4 ± 9.6
Sagittarius	24.8% 574/2,313	144.6 ± 9.4
Pisces	24.1% 566/2,347	173.3 ± 10.3
Virgo	24.1% 555/2,307	162.2 ± 9.9
Aquarius	22.5% 540/2,399	164.8 ± 9.7

Table 4 Correlation between lunar phases and operations for gastrointestinal perforation ($\chi^2 = 1.179$, $P = 0.758$) and lunar phases and operations for perforated aortic aneurysm ($\chi^2 = 2.231$, $P = 0.526$)

Lunar phase	Gastrointestinal perforation	Perforated aortic aneurysm
Waning moon	1.7% 225/13,019	0.4% 57/13,019
New moon	1.4% 13/912	0.2% 2/912
Full moon	2.1% 19/916	0.3% 3/916
Waxing moon	1.8% 231/13,067	0.5% 66/13,067

operations significantly differed from the mean emergency rate during the rest of the evaluation period, but whereas estimated blood loss was comparable (Table 3). A total of 128 severe vascular (0.5 % overall) cases and 448 (1.7% overall) severe gastrointestinal emergency cases were found with a peak for perforated aortic aneurysms in the zodiac sign Aries (0.7%) and a peak for gastrointestinal

Table 5 Association between zodiac signs and operations for gastrointestinal perforation ($P = 0.398$) and operations for perforated aortic aneurysm ($P = 0.562$)

Zodiac sign	Gastrointestinal perforation	Perforated aortic aneurysm
Pisces	1.3% 30/2,347	0.5% 12/2,347
Virgo	1.9% 43/2,307	0.3% 8/2,307
Cancer	1.6% 42/2,572	0.5% 13/2,572
Leo	2.0% 47/2,363	0.4% 10/2,363
Sagittarius	1.3% 30/2,313	0.6% 14/2,313
Scorpio	2.1% 51/2,461	0.5% 12/2,461
Capricorn	2.1% 39/1,877	0.6% 11/1,877
Taurus	2.0% 46/2,307	0.3% 6/2,307
Libra	1.5% 36/2,333	0.6% 13/2,333
Aquarius	1.8% 42/2,399	0.3% 8/2,399
Aries	1.9% 41/2,134	0.7% 14/2,134
Gemini	1.6% 41/2,501	0.3% 7/2,501

perforations in Capricorn and Scorpio (2.1% each); however, neither reached statistical significance. The distribution of both entities was independent of lunar phases and zodiac signs, as shown in Tables 4 and 5.

Discussion

Moon and lunar cycles played an important role in all ancient human civilizations, the earliest reports dating to Sumerian and Egyptian high cultures [14]. The idea that stars and planets may influence human health and behavior can be traced to at least Roman times [15]. Research suggests that a high proportion of health professionals continue to hold this belief [4]. The fear of Friday the 13th is called *Paraskavedekatriaphobia* and was first mentioned by Coriat in 1911 [16]. A number of publications supporting the hypothesis of a lunar influence [17], Friday 13th [11, 18, 19], and zodiac signs alternate with a larger number of publications rejecting any such correlation [3, 8, 10, 20–22]. When analyzing mentally ill people, only the

mental health of patients with schizophrenia has been shown to be affected by moon phases [8]. Kredel et al. report that postoperative nausea and vomiting are not influenced by lunar cycles [20]. The first medical doctor scientifically dealing with the problem of bleeding and lunar phases was Andrews in 1960 [18]. In his original publication, only a pseudo-coincidence of postoperative bleeding complications after tonsillectomy was found [23]. The myth that moon phases influence long term follow-up in oncological surgery was broken by May et al. [22], as well as by Kuehn et al. [21]. As there are no studies evaluating moon effects, zodiac signs, or Friday the 13th on intraoperative blood loss and emergency frequency, we have evaluated almost 28,000 patient records in 111 lunar cycles representing the biggest cohort dealing with this question to our knowledge. When analyzing big cohorts for cyclical aspects or trends—as given by moon phases and zodiac signs—the number of investigated cycles is more important than the overall number of investigated subjects. The results of our study are in line with most previously published studies dealing with the influence of folk belief on surgical complications [3, 23, 24]. The fact that emergency cases are more frequent during the zodiac signs of Capricorn, Aries, and Gemini is easy to explain because the religious Christian holidays (Christmas, Easter, and Whitsun) cumulate in these periods. By definition, all operations performed on holidays were classified as emergencies. It is important to note that the emergency cases investigated in our study are not influenced by human behavior as described in previous studies [11, 19]. This potential bias is unmistakably excluded in our discipline, as emergency cases in general, visceral, and vascular surgical occur only due to internal patient characteristics and therefore represent unpredictable, stochastic events.

Admittedly, some shortcomings of the present study have to be faced. First, one could argue that there may be some systematic error as the data represent only a single-center experience with 3,000 patients treated per year. However, our database comprises all patients treated in general, visceral, and vascular surgery at a tertiary referral centre for 1 million people. Second, our study is limited insofar that the statistical power for both perforated aortic aneurysm and gastrointestinal perforation is 0.125 and 0.203, respectively. To reach a desired power level of 0.8, this study would require 135,025 perforated aortic aneurysm patients and 255,395 gastrointestinal perforation patients. Such large cohorts can only be evaluated in population-based registries [25, 26] but are not available for surgical patients. To our knowledge, however, no study dealing with the problem of superstition in medicine to date has provided a detailed power and sample size calculation. Third, recent studies dealing with the influence of the full moon on medical emergency cases may face a large degree

of beta error because of small patient cohorts and rare events [3, 17, 21, 24, 27]. To overcome the problem of analyzing rare events in a cyclical context within a limited number of patients, some of the recent studies have not clearly defined the event of “full moon” or allocated this event to a time period of several days [22, 27]. Fourth, the estimation of intraoperative blood loss is strongly biased by surgeons’ tendency to underestimate blood loss systematically [28, 29]. At this point, the single-center approach may provide an advantage by reporting standardized measurements and correct documentation of intraoperative blood loss. A final criticism of the present study may be that 128 days are lacking in our investigation because neither emergency cases nor elective surgeries were performed on those days. However, the proportion of lunar phases and zodiac signs in these lacking days was comparable to the rest, so that no statistical bias should occur.

Looking back, numerous studies have dealt with myths and popular belief in modern medicine. Although most publications deny any influence of superstition, some studies still support this view [17, 30], and evidence is weak. Therefore, further research in the context of rare unpredictable events—especially when dealing with cyclic phenomena (e.g., moon phases, weather conditions, seasons)—will require large databases of patients, which can be recruited preferably from multicentric settings. In view of the fact that trauma- and orthopedic surgery have been excluded from the data set reported here, we finally want to state that the impact of lunar phases and other influences on human behavior that would have an impact on risk taking and injury or trauma cannot be answered by the present study.

Taken together, our results support the evidence that operation scheduling according to lunar phases and zodiac signs, as well as avoiding Friday the 13th for elective surgical procedures is nothing more than medical nonsense. Consequently, on the one hand, there is no reason for patients to fear adverse health effects by scheduling operations during full moon phases or on Friday the 13th. On the other hand there is no reason for health professionals in emergency services to avoid Friday the 13th or full moon phases on the staff roster. Finally, we want to highlight that the current investigation is more than a humorous analysis of a popular belief. The possible effect of these factors on operation scheduling would be quite relevant for personal and resource planning.

Conclusions

The hypothesis that lunar phases, zodiac signs, and Friday the 13th influence blood loss and emergency operations is not valid and only myth. So far, however, studies with

adequate power and sample size calculation rejecting any association between superstition and medicine are under-represented in the medical literature.

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Conflict of interest None.

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